Petrie Transport on the Dodecahedron and a V_4-Cover of the Petersen Line Graph

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Abstract

We introduce a 60-vertex tetravalent graph arising from a transport construction on the combinatorial dodecahedron. The resulting graph admits a free V_4 action whose quotient is the line graph of the Petersen graph.

1 Introduction

The combinatorics of regular polyhedra provide a natural source of finite graphs arising from local transport rules on their incidence structures.

Starting from the flag structure of the combinatorial dodecahedron we construct a 60-vertex graph G_{60} using two local transport operations.

The resulting graph is connected, 4-regular, and has 120 edges.

A key feature of the construction is the presence of two commuting involutions generating a free action of the Klein four group

$$V_4 \cong \mathbb{Z}_2^2.$$

The quotient graph

$$Q_{15} = G_{60}/V_4$$

has 15 vertices and degree 4.

We show that the quotient graph coincides with the graph of opposite-edge classes of the dodecahedron and therefore with the line graph of the Petersen graph, a classical object in algebraic graph theory [2].

The graph also appears in the census of tetravalent arc-transitive graphs as `AT4val[60,6]` [3].

The remainder of the paper establishes these connections.

2 Combinatorics of the Dodecahedron

Let D denote the combinatorial dodecahedron.

The dodecahedron has

$$|V| = 20, \quad |E| = 30, \quad |F| = 12$$

with pentagonal faces.

Background on the combinatorics of regular polyhedra may be found in [1].

A *flag* is an incident triple (v, e, f) where

$$v \in e \subset f.$$

Since every edge has two endpoints and two incident faces, the dodecahedron contains

$$120$$

flags.

3 Chamber States

The vertices of the graph constructed in this paper arise from a reduction of the flag set of the dodecahedron.

Recall from Section 2 that the dodecahedron has 120 flags. Each flag may be viewed as a local oriented chamber determined by a vertex, an incident edge, and a face containing that edge.

However, the transport construction we consider does not distinguish between the two opposite orientations of a chamber along an edge. This motivates the following definition.

Definition 1. *Two flags (v, e, f) and (v', e, f') are called* opposite *if v and v' are the endpoints of e and f and f' are the two faces incident to e.*

The unordered pair

$$[(v, e, f)] = \{(v, e, f), (v', e, f')\}$$

is called a chamber state.

Thus a chamber state records the local configuration around an edge of the dodecahedron but ignores the orientation along that edge.

Since each edge contributes exactly four flags and opposite flags are paired, each edge yields two chamber states. Because the dodecahedron has 30 edges, the set of chamber states therefore has size

$$|C(D)| = 60.$$

We denote this set by

$$C(D).$$

These chamber states will serve as the vertices of the graph defined in the next section.

4 Transport Construction

We now define the graph structure on the set of chamber states introduced in the previous section.

Let $C(D)$ denote the set of 60 chamber states of the dodecahedron. Adjacency will be defined using compositions of the local flag operations introduced in Section 2.

4.1 Transport Moves

Let $x = [(v, e, f)]$ be a chamber state represented by a flag (v, e, f).

We consider the following two transport operations:

$$T_1 = FV, \qquad T_2 = VF,$$

where composition is applied right-to-left.

Thus

$$T_1(v, e, f) = F(V(v, e, f)), \qquad T_2(v, e, f) = V(F(v, e, f)).$$

Each transport move produces another flag and therefore determines a new chamber state.

4.2 Definition of the Graph

Definition 2 (Thalean graph)**.** *The* Thalean graph G_{60} *is the graph whose vertex set is* $C(D)$*, where two chamber states* $x, y \in C(D)$ *are adjacent if*

$$y = FV(x) \quad \text{or} \quad y = VF(x),$$

or equivalently if one can be obtained from the other by one application of either transport operation or its inverse.

Because the moves FV and VF are invertible on the flag set, this adjacency relation defines an undirected graph.

Proposition 1. *The graph* G_{60} *is* 4*-regular.*

Proof. From any chamber state x the two transport operations FV and VF produce two neighbors. Their inverses produce two additional neighbors.

These four operations yield four distinct adjacent chamber states, so each vertex has degree 4. □

The graph G_{60} therefore has 60 vertices and degree 4. In the next section we examine the global combinatorial properties of this graph.

5 Basic Graph Properties

We now record the principal combinatorial properties of the graph G_{60} defined in Section 4.

Proposition 2. *The graph* G_{60} *has*

$$|V(G_{60})| = 60, \qquad |E(G_{60})| = 120.$$

Proof. By construction the vertex set is the set $C(D)$ of chamber states of the dodecahedron, which has size 60. Since the graph is 4-regular, the number of edges is

$$|E(G_{60})| = \frac{1}{2} \cdot 60 \cdot 4 = 120.$$

□

Further structural properties of the graph were obtained by direct computation.

figures/g60_graph.png

Figure 1: A visualization of the graph G_{60} obtained from the chamber transport construction.

Proposition 3. *The graph G_{60} contains 40 triangles and has diameter 6.*

To better understand the metric structure of the graph, consider the distance partition from a fixed base vertex. Let $v \in V(G_{60})$ and let

$$S_k(v) = \{u \in V(G_{60}) : d(v, u) = k\}$$

denote the set of vertices at distance k from v.

A computation shows that the shell sizes are

$$(|S_0|, |S_1|, |S_2|, |S_3|, |S_4|, |S_5|, |S_6|) = (1, 4, 8, 16, 24, 6, 1).$$

Thus the graph has diameter 6.

These parameters indicate that G_{60} has a highly structured metric profile rather than that of a typical tetravalent graph. In the next section we show that the graph also admits a natural symmetry coming from two commuting involutions.

6 A V_4 Symmetry of G_{60}

The transport construction admits additional symmetry arising from operations on the chamber states. In this section we describe two commuting involutions acting on the vertex set of G_{60}.

figures/g60_shells.png

Figure 2: Distance layers in the graph G_{60} from a fixed base vertex. The shell sizes are $(1, 4, 8, 16, 24, 6, 1)$, showing that the graph has diameter 6.

6.1 Two Involutions

Let $x = [(v, e, f)]$ be a chamber state.

- The first involution exchanges the two endpoints of the edge e while preserving the chamber state structure.
- The second involution exchanges the two faces incident to the edge e while preserving the endpoints of e.

Each of these operations maps chamber states to chamber states and preserves the adjacency relation defined by the transport moves.

Proposition 4. *Both operations are involutions of the graph G_{60}.*

Proof. Applying either operation twice returns the original chamber state, so each operation has order two. Since the definition of adjacency depends only on the local incidence relations among vertices, edges, and faces of the dodecahedron, the transport moves FV and VF commute with these operations. Therefore each operation preserves adjacency in G_{60}. □

6.2 The Klein Four Action

Proposition 5. *The two involutions commute and generate a group*

$$V_4 \cong \mathbb{Z}_2^2$$

acting on the vertex set of G_{60}.

Proof. Each involution acts independently on a different component of the flag triple. Consequently the two operations commute, and the group they generate is the Klein four group. □

Proposition 6. *The action of* V_4 *on* $V(G_{60})$ *is free.*

Proof. No nontrivial element of V_4 fixes a chamber state, since each operation changes either the vertex or the face component of the flag. Therefore the stabilizer of every vertex is trivial. □

It follows that the vertex set of G_{60} decomposes into orbits of size four under the V_4 action. Since $|V(G_{60})| = 60$, there are exactly

$$\frac{60}{4} = 15$$

orbits.

In the next section we consider the quotient graph obtained from this action.

7 The Quotient Graph

We now consider the graph obtained by taking the quotient of G_{60} under the free action of the Klein four group described in the previous section.

7.1 Definition of the Quotient

Let V_4 act on the vertex set of G_{60}. Since the action is free, each orbit contains exactly four vertices.

Definition 3. *The* quotient graph

$$Q_{15} = G_{60}/V_4$$

is defined as follows.

The vertices of Q_{15} *are the* V_4*-orbits in* $V(G_{60})$*. Two orbits are adjacent if there exist representatives in the corresponding orbits that are adjacent in* G_{60}*.*

Since G_{60} has 60 vertices and the action is free, the quotient graph has

$$|V(Q_{15})| = 15.$$

7.2 Basic Parameters

Proposition 7. *The quotient graph* Q_{15} *is* 4*-regular and has* 30 *edges.*

Proof. Each vertex of G_{60} has degree 4, and the V_4 action preserves adjacency. Therefore each orbit in the quotient inherits four adjacent orbits, so the quotient graph is 4-regular.

Since the graph has 15 vertices and degree 4, the number of edges is

$$|E(Q_{15})| = \frac{1}{2} \cdot 15 \cdot 4 = 30.$$

□

Computational analysis shows that the quotient graph contains 10 triangles and has diameter 3. The shell profile from any vertex is

$$(1, 4, 8, 2).$$

figures/quotient_graph.png

Figure 3: The quotient graph Q_{15} obtained from the free V_4 action on G_{60}.

These parameters suggest that Q_{15} has a natural interpretation in terms of the combinatorics of the dodecahedron. In the next section we describe a graph associated with the opposite-edge classes of the dodecahedron and show that it coincides with the quotient graph.

8 The Opposite-Edge Graph of the Dodecahedron

We now describe a graph naturally associated with the edge structure of the dodecahedron.

8.1 Opposite Edges

Each face of the dodecahedron is a pentagon. In a pentagonal face, two edges are said to be *opposite* if they are separated by one edge on either side of the cycle.

Thus, for a pentagon with cyclic edge order

$$e_1, e_2, e_3, e_4, e_5,$$

the opposite pairs are

$$(e_1, e_3), \quad (e_2, e_4), \quad (e_3, e_5), \quad (e_4, e_1), \quad (e_5, e_2).$$

Taking this relation across the faces of the dodecahedron partitions the 30 edges into 15 opposite-edge classes.

8.2 Definition of the Graph

Definition 4. *Let K_D denote the graph defined as follows.*

- *The vertices of K_D are the 15 opposite-edge classes of the dodecahedron.*
- *Two vertices of K_D are adjacent if the corresponding edge pairs share a common endpoint in the dodecahedron.*

Proposition 8. *The graph K_D has 15 vertices and is 4-regular.*

Proof. Each edge of the dodecahedron meets four other edges at its endpoints. Since each vertex of K_D represents a pair of opposite edges, the adjacency relation preserves this local structure, giving degree 4. □

Consequently,

$$|V(K_D)| = 15, \qquad |E(K_D)| = 30.$$

These parameters match those of the quotient graph Q_{15} obtained in the previous section. In the next section we show that the two graphs are in fact isomorphic.

9 Identification of the Quotient

In this section we identify the quotient graph Q_{15} introduced in Section 7 with the opposite-edge graph K_D defined in Section 8.

9.1 Edge-Based Interpretation of Chamber States

Recall that each chamber state corresponds to an unordered pair of flags associated with a single edge of the dodecahedron. Thus every vertex of G_{60} naturally corresponds to a local configuration around an edge of the polyhedron.

The V_4 action described in Section 6 acts by exchanging endpoints and faces incident to that edge. Consequently, each orbit of the V_4 action consists of the four chamber states determined by the same edge and its opposite orientation choices.

Therefore each orbit corresponds naturally to an *opposite-edge class* of the dodecahedron.

9.2 Correspondence of Vertices

Let

$$\pi : V(G_{60}) \to V(Q_{15})$$

denote the quotient map.

Each vertex of Q_{15} corresponds to a V_4 orbit of chamber states. As noted above, these orbits correspond naturally to the 15 opposite-edge classes of the dodecahedron.

Thus we obtain a bijection

$$V(Q_{15}) \longrightarrow V(K_D).$$

9.3 Adjacency Preservation

It remains to verify that adjacency is preserved under this correspondence.

Two chamber states are adjacent in G_{60} precisely when they are related by one of the transport moves FV or VF. These moves change the edge component of the flag while preserving incidence relations with neighboring vertices and faces of the dodecahedron.

Consequently the transport operations move between chamber states associated with edges that meet at a common vertex of the dodecahedron.

After passing to the quotient under the V_4 action, this adjacency relation becomes exactly the rule defining adjacency in K_D, namely that two opposite-edge classes are adjacent when their edges share a common endpoint.

Theorem 1. *The quotient graph Q_{15} is isomorphic to the opposite-edge graph of the dodecahedron:*

$$Q_{15} \cong K_D.$$

Proof. The correspondence described above defines a bijection between the vertex sets of the two graphs. The preceding discussion shows that adjacency relations coincide under this mapping. Therefore the graphs are isomorphic. □

figures/petersen_line_graph.png

Figure 4: The line graph of the Petersen graph.

In the next section we recall a classical combinatorial fact identifying K_D with the line graph of the Petersen graph.

10 Connection to the Petersen Graph

In this section we complete the identification of the quotient graph obtained in Section 9.

10.1 The Petersen Graph

Recall that the Petersen graph is the 3-regular graph on 10 vertices obtained from the 2-subsets of a 5-element set, where two vertices are adjacent if the corresponding subsets are disjoint. The Petersen graph has 15 edges.

10.2 Line Graph Construction

Given a graph H, the *line graph* $L(H)$ is the graph whose vertices correspond to the edges of H, with two vertices adjacent whenever the corresponding edges share a common endpoint in H.

Thus the line graph of the Petersen graph has

$$|V(L(\text{Petersen}))| = 15.$$

Since each edge of the Petersen graph meets four other edges, the line graph is 4-regular.

10.3 Identification

Proposition 9. *The graph K_D is isomorphic to the line graph of the Petersen graph:*

$$K_D \cong L(\text{Petersen}).$$

Proof. Vertices of K_D correspond to opposite-edge classes of the dodecahedron. These classes correspond naturally to the 15 edges of the Petersen graph.

Under this correspondence, two vertices of K_D are adjacent when the corresponding edge pairs share a vertex of the dodecahedron. This adjacency rule coincides with the definition of adjacency in the line graph $L(\text{Petersen})$, where vertices are adjacent when the corresponding edges share a common endpoint.

Therefore the graphs are isomorphic. □

Combining this identification with the results of the previous section yields the following theorem.

Theorem 2. *The graph G_{60} obtained from the transport construction on dodecahedral chamber states is a regular V_4-cover of the line graph of the Petersen graph:*

$$G_{60} \to L(\text{Petersen}).$$

10.4 Automorphism Group

The Petersen graph has automorphism group

$$\text{Aut}(\text{Petersen}) \cong S_5,$$

which has order 120. Since G_{60} is a regular V_4-cover of $L(\text{Petersen})$, the automorphism group of G_{60} contains the deck group V_4 together with the lifted automorphisms of the base graph.

Proposition 10.

$$|\text{Aut}(G_{60})| = 480.$$

Proof. The deck transformation group contributes a factor of 4, while the automorphisms of the base graph contribute a factor of 120. Thus

$$|\mathrm{Aut}(G_{60})| = 4 \cdot 120 = 480.$$

□

This completes the structural description of the graph G_{60}.

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A Explicit Labeling of the Dodecahedron

For completeness we record an explicit combinatorial model of the dodecahedron used in the computational verification of the results in this paper.

A.1 Vertex Set

The vertex set is

$$V(D) = \{0, 1, \ldots, 19\}.$$

A.2 Edge Set

The 30 edges of the dodecahedron are the following unordered pairs:

$$\begin{aligned} E(D) = \{&(0,1),(1,2),(2,3),(3,4),(4,0),\\ &(0,5),(1,6),(2,7),(3,8),(4,9),\\ &(5,10),(6,11),(7,12),(8,13),(9,14),\\ &(10,11),(11,12),(12,13),(13,14),(14,10),\\ &(5,15),(6,16),(7,17),(8,18),(9,19),\\ &(15,16),(16,17),(17,18),(18,19),(19,15)\}. \end{aligned}$$

A.3 Faces

The 12 pentagonal faces are:

$$
\begin{aligned}
F(D) = \{&(0,1,2,3,4),\\
&(0,5,10,14,9),\\
&(1,6,11,10,5),\\
&(2,7,12,11,6),\\
&(3,8,13,12,7),\\
&(4,9,14,13,8),\\
&(5,6,16,15,10),\\
&(6,7,17,16,11),\\
&(7,8,18,17,12),\\
&(8,9,19,18,13),\\
&(9,5,15,19,14),\\
&(15,16,17,18,19)\}.
\end{aligned}
$$

This labeling determines the full incidence structure of the dodecahedron used in the computations.

B Computational Verification

The structural properties of the graph G_{60} were verified computationally.

B.1 Graph Encoding

The graph was exported in `graph6` format as

`{to?GKC@H??@?BGA@G?A@A?G?Q??CG?_G??????G??@???D???K????????G???B????O???@G???_?????C???OA?????`

This encoding uniquely determines the graph.

B.2 Verified Invariants

Using standard graph algorithms, the following properties were confirmed:

- number of vertices: 60
- number of edges: 120
- degree: 4
- number of triangles: 40
- diameter: 6
- distance shell profile:

$$(1,4,8,16,24,6,1)$$

B.3 Quotient Verification

The action of the Klein four group V_4 generated by the two involutions described in Section 6 was computed explicitly. The vertex set decomposes into 15 orbits of size 4.

The quotient graph was then constructed and verified to satisfy

$$|V| = 15, \qquad |E| = 30, \qquad \deg = 4.$$

Its shell profile was found to be

$$(1, 4, 8, 2).$$

B.4 Identification with $L(\text{Petersen})$

An isomorphism test confirms that the quotient graph is isomorphic to the line graph of the Petersen graph.

B.5 Automorphism Group

The automorphism group of G_{60} was computed and found to have order

$$|\mathrm{Aut}(G_{60})| = 480.$$

This agrees with the theoretical description of G_{60} as a regular V_4-cover of $L(\text{Petersen})$.

References

[1] H. S. M. Coxeter. *Regular Polytopes.* Dover, 1973.

[2] Chris Godsil and Gordon Royle. *Algebraic Graph Theory.* Springer, 2001.

[3] Primož Potočnik, Pablo Spiga, and Gabriel Verret. Census of tetravalent arc-transitive graphs. `https://graphsym.net/`, 2024.